

Tezpur University
Spring Semester Mid Examination-2025

Course Code: CS625

Course Title: Web Technology

Full marks: 30

Time: 90 mins

(The figures in the right-hand margin indicate full marks for the questions)

Q 1. Choose the correct option for the below multiple choice questions. 10X1 = 10 Marks

- (a) QUIC takes 0-RTT _____. (to re-connect to a previously seen client / for sending and receiving the acknowledgement of a packet / for establishing the session key / to connect to a client)
- (b) The Upgrade header in HTTP 1.1 facilitates the hop by hop connection to have _____. (interdependent life cycles / independent data rate / independent compression algorithm / interdependent content type)
- (c) Internet Exchange Point such as NIXI provides better latency by _____. (increasing the Internet bandwidth / deploying many duplicate servers / reducing multiple international hops / collaborating multiple ISPs)
- (d) The repetitive occurrences of same data block in Run Length Encoding algorithm are _____. (replaced with number of occurrences / deleted / appended with the occurrences count / checked for correctness)
- (e) The data transfer connection in FTP _____. (open and closes after each block of a file sent / opens and closes for each file sent / open and closes for each byte sent / open and closes after every time interval set)
- (f) The main disadvantage of using HTTP cookie is _____ (it consume lots of space in the hard drive / it stores the login details / it send the cookies to the server in every request sent / None of the options)
- (g) The field used by the intermediate nodes to indicate the preference list for the acceptance of encoding algorithm while forwarding the message to next intermediate node is _____. (Accept Encoding / TE / Content Encoding / Transfer Encoding)
- (h) Any image file sent along with an e-mail message is opened by the recipient _____ by invoking appropriate application program (MIME protocol / email system / IMAP protocol / POP protocol)
- (i) The security provided in HTTPS is _____. (uni-directional / only for use in HTTP 2.0 / bi-directional / for fast browsers)
- (j) The problem when an application in a recipient computer do not receive the subsequent packets following a packet loss can be resolved by _____. (aggressive re-transmission of the lost packet / opening separate transmission streams / reducing the transmission rate / re-transmission of subsequent packets)

- Q 2. (a) Discuss why FTP Active mode is Server-side initiated. 2 Marks
- (b) Discuss its benefits, the problem and its solution to the problem. 3 Marks
- Q 3. (a) What is Trivial FTP? 2 Marks
- (b) Analyze the presence or absence of error checking and re-transmission resumption from last point mechanisms in FTP and TFTP. 4 Marks
- Q 4. Discuss the statement "IMAP is better as compared to POP in terms of global accessibility but not for server storage space." 4 Marks
- Q 5. Discuss to differentiate between an Internet Exchange point and an Internet Service Provider. 5 Marks

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